Yijia Zhang

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Education

University of Michigan, Ann Arbor - **B.S. in Data Science** - **Bachelor** Sep 2024 - May 2028

Research Interests

Post-training, reasoning RL, RLVR, credit assignment, process supervision, self-improvement, distillation + RL, RL scalling

Research Experience & Publications

AskInfer-Bench: Profiles Are Evidence, Not Intent Labels, in Personalized Deep Research First Author · Under Review at ICLR 2027

- Introduced AskInfer-Bench, a human-grounded benchmark that separates profile conditioning from preference acquisition and evaluation, using controlled RAW0/50/100 persona context masking, selective user disclosure, acquisition traces, and 360 full-factorial deep-research reports across two agent harnesses.
- Showed that persona-derived GPT-5 rubrics achieved 80.8% recall largely through 5.2× overgeneration, collapsing to 39.7% under a matched criterion budget; uncovered diminishing effects from richer user context and a context-dependent clarification reversal—asking helped under cold/partial context (+0.27–0.78) but hurt under full-profile exposure (−0.63–0.74) across three LLM backbones.

What Happens During Autonomous Deep Research After the User Steps Away? Co-First Author · Under Review at ICLR 2027

- Proposed DRaligned, a counterfactual framework for comparing non-aligned long-horizon agent trajectories by isolating single user factors and mapping heterogeneous execution traces into Acquisition, Workspace, and Finalization; designed source-grounded extraction, blinded mirrored evaluation, abstention-aware aggregation, and Matrix/Flow analysis.
- Evaluated 1,691 autonomous agent executions across five model families and multiple harness configurations; found that agents often followed similar research agendas while user-conditioned differentiation emerged much more strongly in final recommendations than acquisition behavior (0.83 vs. 0.15 directional contrast).

Project Experiences (Internship)

DiDi Global Inc. - **Algorithm Intern — Map Business Group** Apr 2026 - Aug 2026

- Developed a language-conditioned LLM agent for pickup-point localization and correction, integrating deterministic geospatial tools such as road binding and cross-road projection for multi-step spatial reasoning; evaluated with acc@10m, acc@link, and online p99 latency, with confidence-aware fallback for uncertain predictions.
- Constructed supervision from real-world GPS drift and pickup-point recommendation failures, generating tool-use trajectories from ground-truth locations and improving performance across heterogeneous scenarios through trajectory balancing and on-policy distillation.
- Built a multimodal geolocation model that predicts fine-grained locations from street-view images under noisy GPS priors, using 160K training examples generated with controlled coordinate perturbations to study robustness to corrupted state information.
- Diagnosed shortcut learning and representation failure through image-only, corrupted-coordinate, and multimodal interventions; explored coordinate corruption, modality dropout, loss reweighting, and token-budget allocation, achieving 69–71% P@30m and 86–94% P@100m on real-world handheld images and analyzing generalization across camera pose, GPS noise, road coverage, and visual landmarks.
- Built a GPT-Image2 → Qwen-Image-Edit-2511 distillation pipeline for large-scale enhancement of low-quality driving-scene annotations and downstream visual understanding of pickup-point environments.

Beijing Langjie Weikang Health Technology Co., Ltd. - **MLE Intern - RAG Knowledge Base Q&A System** May 2025 - Jul 2025

- Developed a medical-domain RAG pipeline using 1.8M consultation records, combining dense retrieval, learned sparse representations, BM25, reciprocal rank fusion, and cross-encoder reranking; achieved over 95% Recall@5 on an internal evaluation set.
- Optimized LLM serving with 4-bit quantization and SGLang, evaluating GPU memory consumption and inference efficiency across deployment configurations.

In-Vehicle Multi-Domain Task-Oriented Dialogue System (LLM + Agent) Nov 2025 - May 2026

- Built an in-vehicle conversational AI system covering 12 domains and 300+ skills, processing 100k+ daily commands.
- Constructed a 500k-sample rejection dataset with DeepSeek pre-labeling and human verification; fine-tuned a 3-layer BERT-Tiny model achieving 90%+ rejection accuracy, millisecond latency, and 500+ QPS.Used DeepSeek-based multi-turn query rewriting, achieving 92% rewriting accuracy across intra-/cross-domain scenarios.
- Built a 400k-sample intent dataset and trained BERT-Large for Top-5 intent recall (Top-1 85%, Top-5 98%)， leveraged Function Calling, slot filling, and MCP and NLG response generation.

Competition Awards

Kaggle Deep Past Challenge - Silver Medal 2026: Rank 115/2673 (Top 5%)

- Built an Ancient Akkadian-English MT system using ByT5-base under low-resource settings (~1500 document-level samples).
- Applied sentence-level alignment, data cleaning, character-level augmentation, and 5-fold cross-validation;Used SWA, dynamic beam search, multi-temperature sampling, and MBR decoding for inference optimization.